

An Atlas of Cross-Context Consistency for 55,000 Drug and Genetic Perturbations

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Abstract

Direction instability (DI) measures whether a perturbation’s effects point the same way across contexts, as the mean pairwise cosine dissimilarity among its signatures. DI is the complement of pairwise phase consistency, applied here at atlas scale: magnitude-corrected, with bootstrap confidence intervals, across $\sim 55,000$ drug and genetic perturbations drawn from LINCS L1000, Tahoe-100M, and the JUMP Cell Painting (JUMP-CP) compound, CRISPR, and ORF datasets.

Eleven hypotheses were pre-registered across three batches before computation. Two met their pre-registered criteria, eight failed, and one gave directional exploratory support. The failures are what constrain the metric: knockout and overexpression DI for the same gene are unrelated ($\rho = 0.011$, $n = 5,220$), and expression variance, paralog count, and paralog expression breadth each explain under 0.4% of DI variance after confound control. The principal positive result is that drug mechanism-of-action class explains 20% of residualized DI variance ($\eta^2 = 0.201$, permutation $p < 0.001$), separating core-machinery inhibitors (mTOR, HDAC; low DI) from receptor-signaling drugs (serotonin and histamine agonists; high DI).

DI computed over JUMP-CP plate replicates indexes technical reproducibility within a single cell line rather than context-dependence across cell types. On those values, essentiality breadth predicts *lower* DI—the opposite of the registered prediction—at partial $\rho = -0.332$, attenuating to $\rho = -0.189$ when signal strength is also controlled; the direction replicates on a two-cell-line CRISPR pilot ($\rho = -0.264$, $n = 128$). The atlas is released as a resource, with per-perturbation DI values, confidence intervals, analysis code, and the frozen pre-registration record.

1 Introduction

High-throughput perturbation screens now profile thousands of drugs and genetic perturbations across panels of cell lines, producing multi-dimensional signatures in transcriptomic [10, 17], morphological [2], and single-cell modalities [21]. Three resources recur throughout: LINCS L1000 (Library of Integrated Network-Based Cellular Signatures, a transcriptomic screen reporting 978 landmark genes), JUMP-CP (Joint Undertaking in Morphological Profiling—Cell Painting, an image-based morphological screen), and Tahoe-100M (a single-cell transcriptomic perturbation atlas). Whether a perturbation’s effects transfer across cellular contexts has practical consequences for drug development and genetic screening [11, 22].

Direction instability (DI) quantifies this context-dependence as $DI = 1 - \overline{\cos(\mathbf{s}_i, \mathbf{s}_j)}$, the mean pairwise cosine dissimilarity among a perturbation’s signatures across contexts.

This quantity is not new. Writing $\bar{R} = \|\sum_i \mathbf{u}_i\|/K$ for the mean resultant length of the unit-normalized signatures, Equation 1 is $\frac{K}{K-1}(1-\bar{R}^2)$, and the mean pairwise cosine is $(K\bar{R}^2-1)/(K-1)$.

The latter is pairwise phase consistency (PPC), introduced by Vinck et al. [20] as a bias-free measure of synchronization—so $DI = 1 - PPC$. PPC was designed for exactly the property claimed here, separating consistency from apparent strength due to sample size. What this study contributes is therefore not the statistic but its application: perturbation-specific, computed across cell lines, magnitude-corrected, and at atlas scale, against a panel of public perturbation datasets that PPC has not been applied to. Related work on perturbation coherence is discussed in Section 3.

DI complements rather than replaces the perturbation-quality and retrieval metrics already in use on these datasets, and the estimands differ. The L1000 Transcriptional Activity Score combines signature strength with replicate concordance to identify transcriptionally active perturbations, so it is designed to register the conjunction that DI separates. Cell Painting percent-replicating asks whether replicate profiles are more similar to each other than a null distribution allows, and mean average precision scores retrieval of known replicate or annotation groups. Each of these answers whether a perturbation produced a reproducible measurable signal, or how well profiles retrieve a stated label. DI answers a different question: given that a signal is present, how much does its direction change across the contexts the analyst designates. A perturbation can score well on activity and retrieval while carrying high DI, and the reverse holds for a weak but directionally stable perturbation. Concurrent work on geometric stability [13, 14] measures perturbation coherence on a related axis; it aggregates over individual cells within one context, whereas DI aggregates across the contexts an analyst designates, so the two quantify different variation and are not interchangeable. Because cosine normalizes each signature to unit length before comparison, DI is immune to the first-order magnitude confound—systematic covariation between effect size and metric value—that dominates alternative geometric measures such as Grassmannian geodesic distance, profile variability, and optimal transport cost [22]. A residual second-order correlation persists near zero (noise-dominated signatures inflate pairwise cosine dissimilarity), which we remove by magnitude correction (Section 1). A companion study [22], a preprint under peer review, reports this immunity. In leave-one-cell-line-out analysis of 2,694 drugs profiled in at least ten cell lines, DI predicted a cosine-defined held-out consistency outcome with AUROC = 0.986 before and 0.864 after magnitude correction. Predictor and outcome share cosine geometry, and a geometry-matched synthetic null in that study reaches AUROC = 0.996, above the value obtained on real drugs; the held-out prediction therefore characterizes out-of-sample geometric consistency rather than validating biological transport. The biological evidence comes instead from criteria that do not involve cosine similarity: Jaccard overlap of held-out top gene sets (AUROC = 0.957), breadth of target gene expression across tissues in CCLE ($\rho = -0.170$, $p = 1.1 \times 10^{-5}$), and PRISM cell-viability variability ($\rho = 0.261$, $p = 0.019$). That study is a preprint under peer review rather than published work, so its headline evidence is reproduced here (Table 1) and this manuscript does not rest on it. Sections that depend on it are marked at the point of use. The present study asks a different question: what does DI capture, and what does it fail to capture, across modalities and perturbation types?

We constructed a DI atlas spanning three measurement technologies and four perturbation classes: drug perturbations measured by L1000 transcriptomics (LINCS, 8,949 compounds) and single-cell transcriptomics (Tahoe-100M, 379 compounds); genetic knockouts measured by Cell Painting morphology (JUMP-CP CRISPR, 7,946 genes); and genetic overexpressions measured by Cell Painting morphology (JUMP-CP ORF, 12,590 genes). For each perturbation, the atlas provides magnitude-corrected DI with bootstrap confidence intervals, enabling downstream analyses that require a calibrated measure of context-dependence.

To characterize this resource, we pre-registered eleven primary hypotheses across three batches before computing DI on new datasets (Batch 1 SHA 0d07a01; Batch 2A SHA a17c125; Batch 3 SHA

Table 1: Validation evidence for DI reported in the companion study, reproduced here so this manuscript can be read on its own. The first row is a geometric characterization rather than biological validation: predictor and outcome share cosine geometry, and a geometry-matched synthetic null reaches AUROC = 0.996 in that study, above the value obtained on real drugs. The remaining three rows do not involve cosine similarity, and the last two are measured on external datasets. The held-out cosine analysis covers 2,694 drugs profiled in at least ten cell lines; the gene-set criterion is computed on LINCS signatures. The companion study is a preprint under peer review at the time of writing.

Channel	Criterion	Effect
Held-out cosine	Leave-one-cell-line-out consistency	AUROC = 0.986 raw, 0.864 corrected
Gene-set	Jaccard overlap of held-out top genes	AUROC = 0.957
Expression	CCLE target-gene breadth across tissues	$\rho = -0.170$, $p = 1.1 \times 10^{-5}$
Viability	PRISM cell-viability variability	$\rho = 0.261$, $p = 0.019$

4c34699). The hypotheses test whether DI is concordant across modalities, whether it correlates with gene essentiality, whether it is an intrinsic property of genes, whether expression variance, drug selectivity, or paralog buffering can explain it, and whether DI is structured by drug mechanism-of-action class. A third plan, frozen after the results of the first two were known, adds three hypotheses: mechanism-of-action stratification, cross-cell-line replication of the essentiality relationship, and a convergent test on DepMap dependency-score variability. Of the eleven primary tests, two met their pre-registered criteria and eight failed; the eleventh, Experiment G, gave directional exploratory support rather than a pass. A twelfth test—PRISM viability concordance—is exploratory.

The results constrain what DI measures. DI carries a modest cross-modal component, evidence that it is not an artifact of one measurement platform: compounds whose transcriptomic effects vary across cell lines also produce variable morphological profiles. Essential genes show *low* DI, against the registered prediction, and the association survives adjustment for signal strength at $\rho = -0.189$; functional constraint producing stereotyped disruption regardless of cellular identity is one hypothesis that would account for it. The sign-flip directionally replicates on true cross-cell-line CRISPR data ($\rho = -0.26$, cpg0000 pilot, $K = 2$ cell lines). Knockout and overexpression DI are unrelated, so DI does not index a stable property of the target gene. Expression variance, paralog count, and paralog expression breadth produce correlations too small to matter biologically (partial $\rho < 0.07$ after confound control), ruling them out as explanations for DI variation. DI varies systematically by drug mechanism-of-action class: core-machinery inhibitors (mTOR, HDAC) show low DI while receptor-signaling drugs (serotonin agonists, histamine agonists) show high DI.

Taken together, these results constrain DI to a dimension of perturbation biology that existing gene annotations do not index, and the atlas makes that measurement available at scale.

The results are reported in four parts. Section 2.1 asks whether DI reflects a property of perturbations rather than of any one measurement platform. Section 2.2 reports the one relationship that holds: mechanism-of-action class structures DI. Section 2.3 reports the gene-level properties that do not explain it. Section 2.4 tests the essentiality relationship on true cross-cell-line data and shows where a scalar operationalization of context-dependence diverges from the vector one.

Materials and methods

The atlas

Datasets

The atlas draws on five perturbation datasets spanning three measurement technologies (Table 2), whose DI distributions are shown in Figure 1. Two drug datasets were previously analyzed in the companion confound study [22]: LINCS L1000 transcriptomics (978 landmark genes, 8,949 compounds across variable numbers of cell lines; 17) and JUMP-CP Cell Painting compound morphology (3,180 features, 25,254 compounds; 3). Three datasets are new to this study: Tahoe-100M single-cell transcriptomics (379 drugs across 47–50 cell lines per drug; 21), JUMP-CP CRISPR knockouts (7,946 genes, 259 post-sphering PCA features, plate replicates as contexts; 4), and JUMP-CP ORF overexpressions (12,590 genes, 722 features, plate replicates as contexts; 4). Together, the atlas covers $\sim 55,000$ perturbations (8,949 LINCS drugs, 379 Tahoe drugs, 25,254 JUMP-CP compounds, 7,946 CRISPR knockouts, 12,590 ORF overexpressions; drug-dataset overlaps are small, $n \leq 378$).

An important asymmetry: LINCS and Tahoe compute DI over cell lines, measuring biological context-dependence, while JUMP-CP CRISPR and ORF compute DI over plate replicates within a single cell line, measuring technical reproducibility. The two concordance tests (Section 2.1) compare datasets that share the same context type (cell lines), but the essentiality, mechanism-specificity, and proxy tests (Section 2.3) use plate-replicate DI. Section 3 discusses how this distinction affects interpretation.

Table 2: Datasets in the DI atlas. “Contexts” denotes the unit over which DI is computed: cell lines for drug datasets, plate replicates for genetic perturbation datasets. Only perturbations with ≥ 5 contexts are included. Both genetic counts differ from the frozen analysis plan’s estimates: it projected 7,948 CRISPR genes and 15,121 ORF genes with ≥ 5 replicates, against actual post-filtering counts of 7,946 and 12,590. The plan’s figures were pre-analysis estimates from the Cell Painting Gallery manifests; the difference is deterministic, arising from quality filtering, and is recorded among the deviations.

Dataset	Modality	Perturbations	Features	Contexts	Source
LINCS L1000	Transcriptomic	8,949 drugs	978	Cell lines	Pre-computed
Tahoe-100M	Transcriptomic	379 drugs	$\sim 62,700$	Cell lines	HuggingFace
JUMP-CP compounds	Morphological	25,254 compounds	3,180	Varied [†]	Pre-computed
JUMP-CP CRISPR	Morphological	7,946 genes	259	Plate reps	CPG
JUMP-CP ORF	Morphological	12,590 genes	722	Plate reps	CPG

CPG = Cell Painting Gallery (cpg0016). [†]Context definitions vary by compound; see companion study [22].

A sixth dataset, PRISM repurposing viability [5], is included as a complementary validation source. PRISM measures scalar cell viability (log-fold-change) rather than high-dimensional profiles, so cosine-based DI is undefined. We compute the coefficient of variation (CV) of viability across cell lines as a scalar analogue of context-dependence. PRISM results are labeled exploratory throughout.

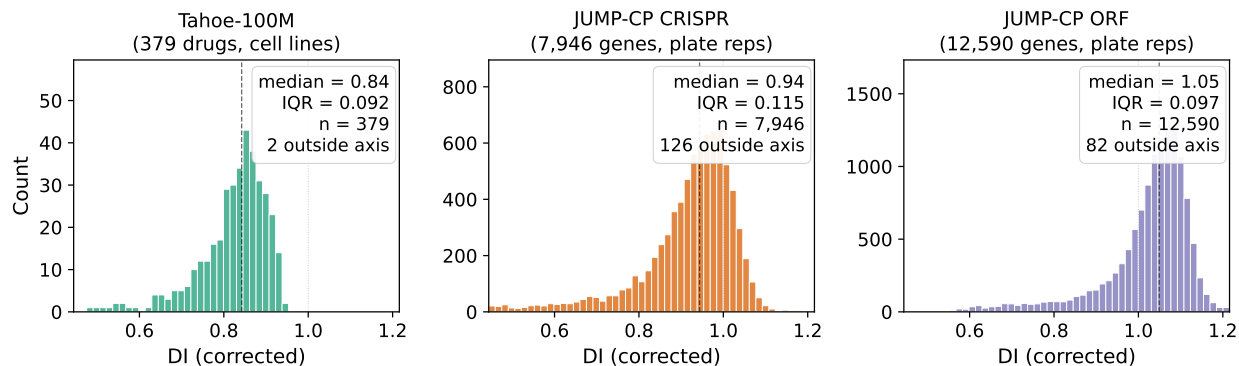


Figure 1: Distribution of magnitude-corrected DI across the three new atlas datasets. Dashed lines mark medians; dotted lines mark $DI = 1$ (orthogonal signatures). DI can exceed 1 when signatures are anti-correlated across contexts. All three panels share an x-axis spanning the 0.5th–99.5th percentile of the pooled values; the count of perturbations falling outside that window is stated in each panel. Tahoe-100M drug DI has the lowest median (0.84, against CRISPR 0.94 and ORF 1.05) and the narrowest spread of the three on every measure: interquartile range 0.092 against 0.115 and 0.097, standard deviation 0.087 against 0.131 and 0.113. The narrower Tahoe distribution is expected from estimator variance rather than from biology: DI is computed from K contexts, and Tahoe has a median K of 49 against 5 for CRISPR and ORF. CRISPR and ORF distributions concentrate near 1 (near-orthogonal signatures across plate replicates), reflecting the high baseline noise of morphological profiling at plate-replicate resolution.

Direction instability

For a perturbation with $K \geq 5$ context-specific signatures $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^d$, direction instability is

$$DI = 1 - \frac{2}{K(K-1)} \sum_{i < j} \frac{\mathbf{s}_i \cdot \mathbf{s}_j}{\|\mathbf{s}_i\| \|\mathbf{s}_j\|}. \quad (1)$$

$DI = 0$ when all signatures point in the same direction (perfectly consistent perturbation), equals 1 when signatures are orthogonal, and exceeds 1 when signatures are anti-correlated. The excess is bounded: K unit vectors cannot be mutually anti-correlated, so $0 \leq DI \leq K/(K-1)$, giving a maximum of 2.000 at $K = 2$, 1.250 at $K = 5$, and 1.021 at $K = 49$. For the Tahoe panel ($K = 49$) the maximum possible excess over 1 is therefore 0.021, and “exceeds 1” should be read quantitatively rather than as room for large anti-correlation. Note also that $DI_{\text{corrected}}$ (Equation 2) is a residual plus an intercept and is not bounded by $K/(K-1)$; it is not interpretable as $1 - \overline{\cos}$, and Figure 1 plots the corrected values. By construction, DI is a reparameterization of mean pairwise cosine similarity (Pearson $r = -1.0$ between DI and $\overline{\cos}$).

Magnitude immunity and correction

The first-order magnitude confound arises when a metric’s expected value changes with effect size: stronger perturbations mechanically produce larger distances (or smaller ones, depending on the metric). Cosine similarity is ratio-scale— $\cos(\alpha \mathbf{s}_i, \alpha \mathbf{s}_j) = \cos(\mathbf{s}_i, \mathbf{s}_j)$ for any $\alpha > 0$ —so DI is invariant to uniform rescaling of all signatures. This is the sense in which DI is “immune by construction”: the dominant confound pathway (stronger effect $\rightarrow$ larger norm $\rightarrow$ larger distance) is algebraically cancelled. The companion study [22] verified this empirically across three modalities (LINCS,

DepMap CRISPR, Perturb-seq), showing that alternative metrics exhibit Cohen’s $d = 0.55$ – 2.65 magnitude confounding while DI shows none after correction.

A residual second-order correlation remains. Weakly-acting perturbations produce noisier signature estimates, and noise inflates the average pairwise cosine dissimilarity even after unit-length normalization. We remove this residual confound by OLS regression of raw DI on mean signature L_2 norm:

$$\text{DI}_{\text{corrected}} = \text{DI}_{\text{raw}} - \hat{f}(\|\mathbf{s}\|) + \hat{\beta}_0, \quad (2)$$

where $\hat{f}$ is the OLS prediction and $\hat{\beta}_0$ is the intercept. This correction is applied per dataset. $\text{DI}_{\text{corrected}}$ is the primary metric throughout; DI_{raw} is reported for transparency.

What the correction leaves behind. $\hat{f}$ is a global linear fit on the untransformed norm, and the dependence it removes is only the linear part of a monotonic, nonlinear relationship. The fit itself explains $R^2 = 0.081$ of raw DI in CRISPR (Appendix E), while a linear model of log mean norm and magnitude CV explains $R^2 = 0.478$ of the *corrected* values in the same dataset. Magnitude-related structure therefore survives the correction in CRISPR, and analyses on CRISPR DI control for norm and magnitude CV explicitly rather than relying on the correction alone (Section 2.3). The residual is smaller in ORF ($R^2 = 0.313$) and small in Tahoe ($R^2 = 0.076$), consistent with Tahoe’s higher signal-to-noise ratio.

Bootstrap confidence intervals

For each perturbation, we resample the K contexts with replacement 1,000 times and recompute DI on each bootstrap sample. The 2.5th and 97.5th percentiles define the 95% bootstrap confidence interval. These intervals quantify the stability of each DI estimate with respect to the particular set of contexts observed.

Statistical framework

All hypothesis tests use partial Spearman correlation with confound control. Spearman rather than Pearson correlation is used throughout because the registered hypotheses are monotonic rather than linear, because DI and several covariates (essentiality breadth, paralog count, cell-line count) are bounded or heavily skewed, and because rank statistics limit the leverage of the extreme $\text{di}_{\text{corrected}}$ values that the residual-plus-intercept definition admits. The partial correlation is computed by Pearson-residualizing both variables against the covariate matrix (with intercept column), then computing Spearman’s ρ on the residuals. This method avoids the bias of rank-then-residualize approaches, which produce spurious partial correlations of $\rho \approx 0.05$ – 0.25 under the null at large n (verified in simulation; see pre-registration documents). Confidence intervals use the Fisher z -transform with degrees of freedom adjusted for the number of covariates.

Where the predictor is categorical rather than continuous, effect size is reported as partial eta-squared from a one-way ANOVA,

$$\eta_p^2 = \frac{SS_{\text{between}}}{SS_{\text{between}} + SS_{\text{within}}}, \quad (3)$$

which is the proportion of variance in the residualized outcome attributable to class membership. It replaces ρ for the mechanism-of-action test because mechanism-of-action class is unordered, so no monotonic relationship is defined and a rank correlation would not be interpretable. Degrees of freedom are reported with the ANOVA, and significance is assessed by permutation of class

labels rather than against the F reference distribution, since the class sizes are unequal and the residualized outcome is not normal.

Every primary test applies two decision criteria simultaneously: (i) $p < \alpha$ (Bonferroni-corrected: $\alpha = 0.0125$ for Batch 1’s four primary tests, $\alpha = 0.0167$ for Batch 2A’s three hypotheses and Batch 3’s three), and (ii) the 95% CI lower bound exceeds ± 0.05 in the predicted direction. The CI floor is necessary because at $n > 5,000$, effects as small as $\rho = 0.03$ reach statistical significance. Without the floor, trivially small correlations would pass.

Data format

The atlas is distributed as CSV files containing, for each perturbation: DI_{raw} , $DI_{\text{corrected}}$, bootstrap CI bounds, magnitude CV, mean signature norm, and number of contexts. Full data provenance and repository locations are listed in the Data Availability section.

Analysis plans and deviations

All hypotheses, decision criteria, statistical methods, and analysis code were frozen before execution. Batch 1 (concordance, essentiality, knockout vs. overexpression, PRISM) was frozen at commit SHA 0d07a01; Batch 2A (expression variance, drug selectivity, paralog count, paralog breadth) at commit SHA a17c125; Batch 3 (MoA stratification, cross-cell-line CRISPR, DepMap convergent) at commit SHA 4c34699. The pre-registration documents specify the statistical method, Bonferroni correction levels, decision criteria, power analyses, and interpretation guides for each outcome. All three documents are included in the repository.

The three plans were not written at the same time. The first two were frozen before any result in this study was known. The third was written after the results of the first two were known, and specifies three further hypotheses: mechanism-of-action stratification, cross-cell-line replication of the essentiality relationship, and a convergent test on dependency scores. Its hypotheses are prespecified with respect to their own outcomes and post hoc with respect to the earlier results, and are marked as such wherever they are reported. Freeze times are verifiable against the repository history.

Two further disclosures attach to that third plan. The mechanism-of-action analysis is exploratory-then-confirmed: a raw, uncontrolled $\eta^2 \approx 0.42$ was seen during scoping before the plan was frozen, and the confirmatory confound-controlled, permutation-tested version was unseen at freeze time. For the cross-cell-line replication, the analysis logic is byte-identical between the frozen commit and the executed version; only the data download function was changed after freezing, to correct S3 paths that returned 404 errors. Both are documented in the repository.

The confidence-interval floor

At sample sizes above $n = 5,000$, trivially small effects reach statistical significance. The pre-registration introduced a uniform CI lower-bound gate ($|CI \text{ bound}| > 0.05$ in the predicted direction) to prevent this. The gate blocked expression variance ($\rho = 0.060$), which would have passed on p -value alone. Paralog breadth ($\rho = -0.029$) fails on both criteria: its $p = 0.017$ marginally exceeds the Bonferroni threshold and its CI does not clear the floor. In both cases, the effects explain less than 0.4% of variance and lack biological plausibility as primary explanations for DI variation.

Deviations

Seven deviations from the pre-registration occurred during analysis. All are data-wrangling corrections or sample-size discrepancies; none alter the statistical method, decision criteria, or interpretation framework.

DepMap version. All DepMap analyses use the pre-registered 24Q4 release (figshare article 27993248): Chronos dependency scores (file 51064667; 1,178 cell lines, 17,916 genes) for the essentiality test, expression (file 51065489; 1,673 cell lines, 19,193 genes) for expression variance and paralog tests. Initial runs inadvertently used 22Q2 Chronos (essentiality) and 24Q2 expression (expression variance, paralogs). Re-running on the correct 24Q4 release changed no verdicts: the essentiality test moved from $\rho = -0.330$ to -0.332 , expression variance from 0.064 to 0.060, and paralog count/breadth were identical to three decimal places.

Same-modality concordance matching method. The pre-registration specified InChIKey matching for drug concordance. An initial run erroneously used case-insensitive drug-name matching ($n = 114$, $\rho = -0.003$), which introduced approximately 38 spurious pairs from salt forms, stereoisomers, and name collisions. Reverting to InChIKey matching via PubChem CID cross-reference ($n = 76$, $\rho = 0.254$) moved toward the pre-registered method, resolving the spurious null. The InChIKey result is the pre-registered analysis; the name-matched result is documented in Appendix D.

Batches 1–2A deviations. (i) Cross-modal concordance yielded $n = 378$ instead of the estimated $n = 330$, because `pert_iname` matching recovered more valid InChIKey pairs than anticipated. (ii) `pert_iname` was used instead of `pert_id` for LINCS metadata lookup; `pert_id` matching was a data-wrangling error yielding only 8 compounds. (iii) Context definitions differ across datasets (cell lines vs. plate replicates); see Section 1 and Section 3. (iv) Drug selectivity yielded $n = 75$ instead of the estimated $n \sim 1,000$ due to the small LINCS-PRISM overlap after viability filtering. (v) Nine genes were dropped from the paralog tests due to missing expression values, leaving $n = 6,894$ (paralog count) and $n = 6,882$ (paralog breadth). (vi) The CRISPR and ORF panels contain 7,946 and 12,590 perturbations against the plan’s pre-analysis estimates of 7,948 and 15,121; both differences follow deterministically from the quality filter and neither affects a decision criterion.

Batch 3 deviations. (i) Experiment F: The scoping disclosure (raw η^2 observed before freeze) is described above. The confirmatory analysis (confound-controlled, permutation-tested) was unseen at freeze time. (ii) Experiment G: The pre-registration estimated $n \approx 160$ shared genes; the actual overlap with DepMap is $n = 128$ (deterministic: genes present in both cell lines with non-zero signature norms). The data download function was modified post-freeze to use the correct S3 paths; the analysis logic (DI computation, correlation, CI) is byte-identical to the frozen version. (iii) After obtaining the Experiment G result, bootstrap and timepoint-stratified sensitivity analyses were run exploratorily. These were not pre-registered and are excluded from the manuscript (documented in `DEVIATIONS.md`).

2 Results

2.1 Concordance across measurement modalities

The atlas is useful only if DI reflects a biological property of perturbations rather than a measurement artifact of any single platform. Two pre-registered concordance tests assess this by comparing DI values for the same compounds measured in different datasets. An exploratory check against PRISM cell viability, a scalar outcome with a different noise structure, is reported in Appendix B.

2.1.1 Transcriptomic against morphological

Compounds present in both LINCS L1000 (transcriptomic DI) and JUMP-CP Cell Painting (morphological DI) were matched by InChIKey connectivity prefix (14 characters). Of the matched set, $n = 378$ compounds have ≥ 5 contexts in both datasets.

Partial Spearman $\rho = 0.190$ (95% CI [0.090, 0.285], $p = 2.1 \times 10^{-4}$), controlling for LINCS n_{contexts} as one covariate. The CI lower bound (0.090) exceeds the 0.05 floor, and $p < 0.0125$. This test passes both pre-registered criteria.

The effect size falls in the “weakly concordant” bin ($0.15 \leq \rho < 0.30$) defined in the pre-registration, consistent with the expectation that cross-modal comparisons capture only the fraction of biology shared between gene expression and cell morphology [8]. The concordance is weak but genuine: compounds whose transcriptomic signatures vary across cell lines also produce variable morphological profiles, and this relationship survives confound control.

2.1.2 Transcriptomic against transcriptomic

Drugs present in both LINCS L1000 and Tahoe-100M were matched by InChIKey connectivity prefix, verified via PubChem CID cross-reference. This yielded $n = 76$ overlapping compounds (fewer than the pre-registered estimate of $n \approx 103$, because some Tahoe drugs lack PubChem CID annotations mapping to LINCS InChIKeys). LINCS DI is computed in $\mathbb{R}^{978}$ (landmark genes) while Tahoe DI is computed in $\mathbb{R}^{\sim 62,000}$ (full transcriptome). Because cosine similarity is invariant to positive rescaling, DI values are comparable across feature spaces of different dimensionality.

Partial Spearman $\rho = 0.254$ (95% CI [0.027, 0.456], $p = 0.029$), controlling for LINCS and Tahoe n_{contexts} (two covariates). This test **fails**: the CI lower bound (0.027) does not clear the 0.05 floor, and $p = 0.029$ exceeds the Bonferroni threshold ($\alpha = 0.0125$).

The failure is one of statistical power, not absence of concordance. The point estimate ($\rho = 0.254$) is larger than the cross-modal result ($\rho = 0.190$), and the pre-registered power analysis predicted this outcome: at $n = 76$, the study achieves 80% power only for true effects ≥ 0.35 . We interpret this pattern as a sample-size artifact—the LINCS-Tahoe overlap ($n = 76$) is far smaller than LINCS-JUMP ($n = 378$)—rather than evidence that DI transfers better across modalities than within one. A deviation in the matching method is disclosed in Section 1.

2.2 Mechanism-of-action structure

Scoping disclosure. A raw, uncontrolled one-way ANOVA of DI on mechanism-of-action class was computed during scoping and gave $\eta^2 \approx 0.42$; that number was seen before the analysis plan was frozen. The confirmatory test specified in the plan is the confound-controlled, permutation-tested version reported below, whose result was unseen at freeze time. This analysis is therefore exploratory-then-confirmed.

Batches 1 and 2A established which gene-level properties fail to explain DI. The natural follow-up is whether DI varies systematically with drug mechanism-of-action class. If core-machinery inhibitors (targeting fundamental cellular processes) show low DI while receptor-signaling drugs (engaging cell-type-specific pathways) show high DI, then DI captures biologically interpretable structure, and the atlas stratifies drug classes in a useful way.

Because DI depends mechanically on profiling depth (number of cell lines) and MoA classes may differ in profiling depth, the primary analysis uses residualized DI: raw DI residualized on $n_{\text{cell lines}}$ and mean signature norm via OLS. One-way ANOVA of residualized DI on MoA class (80 classes with ≥ 5 drugs, 1,031 drugs total; $\text{df} = 79, 951$) yields partial $\eta^2 = 0.201$. The raw (uncontrolled) η^2 is 0.417; controlling for profiling depth and signal strength halves the effect, confirming that the confound control is load-bearing.

A 1,000-permutation null (shuffling MoA labels, preserving class sizes) produced a null distribution with mean = 0.077, SD = 0.012, 99th percentile = 0.111. The observed $\eta^2 = 0.201$ exceeds the 99th percentile and the empirical permutation $p < 0.001$. Both pre-registered criteria are met.

The biological pattern is interpretable. The lowest-DI MoA classes are mTOR inhibitors, HDAC inhibitors, HSP inhibitors, and PI3K inhibitors—drugs targeting core cellular machinery [16] that operates uniformly across cell types. The highest-DI classes are serotonin receptor agonists, histamine receptor agonists, GABA receptor antagonists, and PPAR receptor agonists—drugs acting through receptor-mediated signaling pathways whose expression and coupling vary across cell lineages.

Mechanism-of-action class therefore accounts for a fifth of residualized DI variance, and the classes separate along a pharmacologically interpretable axis rather than an arbitrary one.

2.3 Gene-level properties that do not explain DI

Five pre-registered experiments (Batches 1 and 2A) tested whether DI variation across genes and drugs can be explained by simpler biological properties. All five fail their pre-registered criteria, and each failure constrains the interpretation of DI in a specific way. Figure 2 shows each effect against the ± 0.05 confidence-interval floor, which is what separates the failures from the two passes. Table 3 summarizes the full scorecard including the replication and divergence tests (Section 2.4).

2.3.1 Essentiality breadth

The pre-registration predicted that broadly essential genes would show high morphological DI: genes required in many cell types should produce context-dependent phenotypic disruption. The observed correlation is large, precisely estimated, and in the opposite direction. Partial Spearman $\rho = -0.332$ (95% CI $[-0.352, -0.312]$, $p \approx 0$, $n = 7,653$), controlling for n_{contexts} . Essentiality breadth (fraction of DepMap 24Q4 cell lines with Chronos score < -0.5 ; 6, 7, 19) predicts *lower* DI, accounting for approximately 11% of variance.

Attenuation under signal-strength control. Because CRISPR DI is computed across plate replicates, a purely measurement-side pathway predicts the same sign: an essential gene produces a large phenotype, a large phenotype gives a higher signal-to-noise ratio, replicate signatures then agree more closely, and DI falls. That pathway requires no appeal to functional constraint. It is present in these data: essentiality breadth correlates with mean signature norm at Spearman $\rho = +0.259$, and corrected DI retains structure related to signal strength ($\rho = -0.212$ with mean norm, $\rho = +0.279$ with magnitude CV).

Table 3: Pre-registered experiment results. Batches 1–2A use partial Spearman ρ with confound control ($\alpha = 0.0125$ for Batch 1, 0.0167 for Batch 2A) except the KO vs. OE test (raw Spearman + permutation null). Batch 3 ($\alpha = 0.0167$, three tests) was prompted by peer review; Experiment F reports partial η^2 instead of ρ . “CI gate” indicates that the 95% CI lower bound did not clear the ± 0.05 threshold in the predicted direction. [‡]Reported in Appendix B: uncorrected p , no covariate adjustment, $n = 75$.

Batch	Hypothesis	n	Effect	p	Verdict
1	Cross-modal concordance	378	$\rho = 0.190$	2.1×10^{-4}	PASS
1	Same-modality concordance	76	$\rho = 0.254$	0.029	FAIL (CI gate)
1	Essentiality $\rightarrow$ high DI	7,653	$\rho = -0.332$	~ 0	FAIL (wrong sign)
1	KO $\approx$ OE DI	5,220	$\rho = 0.011$	0.43	FAIL (null)
2A	Expression variance	7,802	$\rho = 0.060$	1.1×10^{-7}	FAIL (CI gate)
2A	Drug selectivity	75	$\rho = 0.144$	0.22	FAIL (underpowered)
2A	Paralog count	6,894	$\rho = -0.007$	0.57	FAIL (null)
2A	Paralog breadth	6,882	$\rho = -0.029$	0.017	FAIL (CI gate + p)
3	MoA stratification (F)	1,031	$\eta^2 = 0.201$	< 0.001	PASS
3	Cross-cell-line ess. (G)	128	$\rho = -0.264$	0.003	Directional (exploratory)
3	DepMap convergent (H)	17,916	$\rho = +0.811$	~ 0	FAIL (wrong sign)
<i>Exploratory</i>					
1	Viability concordance [‡]	75	$\rho = 0.359$	0.0016	Exploratory

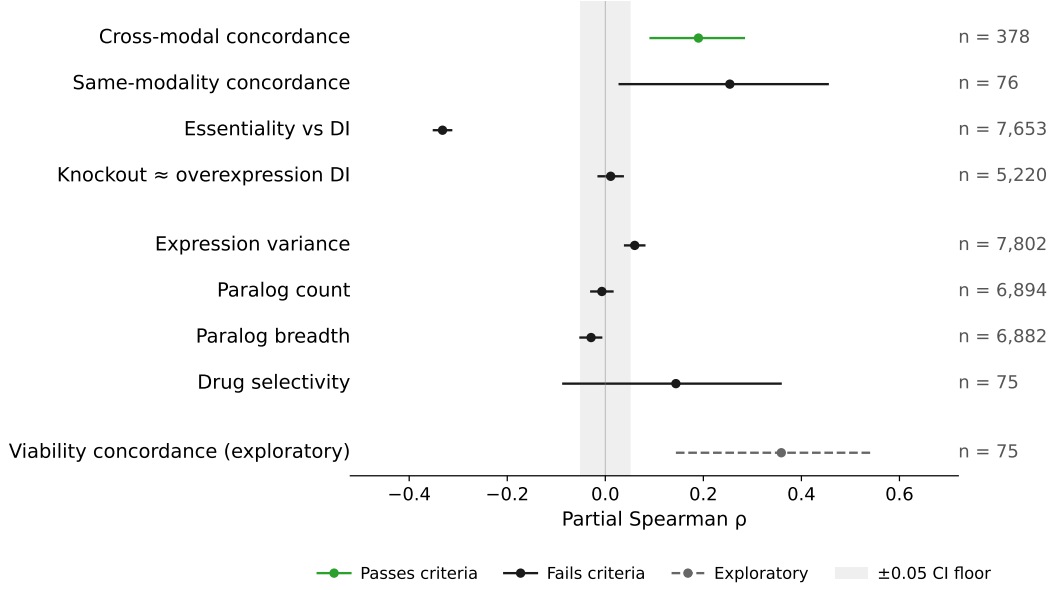


Figure 2: Pre-registered experiment results (Batches 1–2A). Each point shows the partial Spearman ρ (or raw Spearman for the knockout vs. overexpression test) with 95% confidence intervals. The gray band marks the ± 0.05 CI floor: a test passes only if its CI lower bound clears this band in the predicted direction. Same-modality concordance has a larger point estimate ($\rho = 0.25$) than cross-modal ($\rho = 0.19$) but fails because its CI lower bound (0.027) falls inside the floor at $n = 76$. Only cross-modal concordance passes both pre-registered criteria; the seven primary failures delimit what DI measures. The mechanism-of-action, replication and divergence results are reported in Sections 2.2 and 2.4 and Table 3.

Adding mean norm to the covariate set moves the partial correlation from -0.332 to -0.243 ; adding magnitude CV as well moves it to $\rho = -0.189$ (95% CI $[-0.211, -0.168]$, $p \approx 0$, $n = 7,653$), a reduction of roughly 43%. **We report $\rho = -0.189$ as the effect after signal-strength control, and treat it rather than -0.332 as the quantity the functional-constraint reading has to explain.** Roughly half of the raw association is attributable to signal strength in a metric that is magnitude-invariant by construction, which is itself a constraint on how far that construction protects against magnitude.

Mean norm may be a mediator rather than a confounder here, since knocking out an essential gene should produce a large phenotype. The design cannot separate the two, and that ambiguity is the reason the residual association is reported as a bound rather than as an estimate of functional constraint.

Figure 3 shows the relationship. The sign-flip admits a natural interpretation, which we emphasize is *post-hoc*: the pre-registration confused two senses of context-dependence. Essential genes affect many cell types, but they affect them in the same way. A gene required for mitosis, ribosome assembly, or DNA repair produces a stereotyped catastrophe regardless of cellular identity. Functional constraint makes the downstream phenotypic consequences uniform across contexts, yielding low DI. Genes with high DI tend to be selectively essential or non-essential—their knockout phenotypes interact with cell-type-specific pathways, producing variable consequences across the panel.

This interpretation generates a falsifiable prediction: among non-essential genes, those with

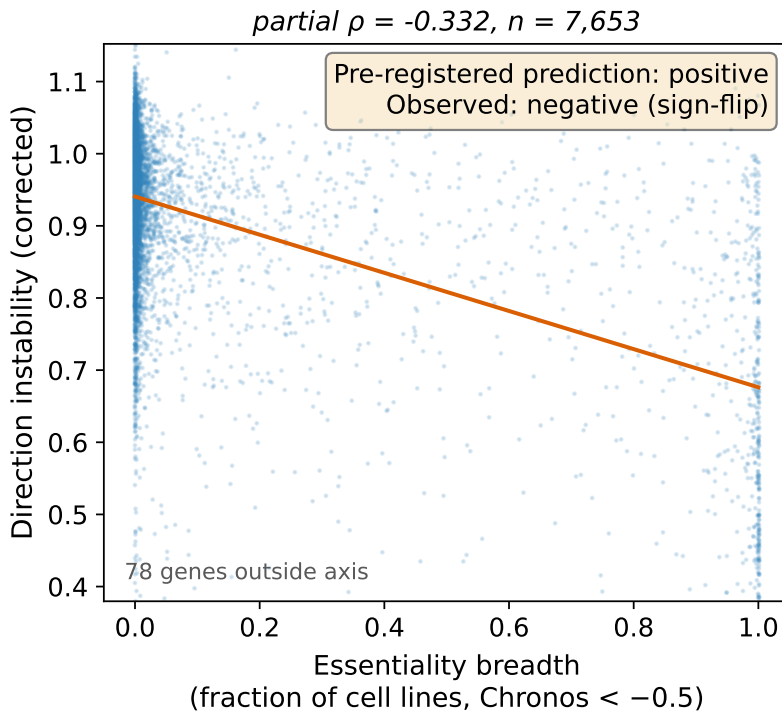


Figure 3: Essentiality breadth (fraction of DepMap cell lines with Chronos < -0.5) vs. magnitude-corrected DI for $n = 7,653$ JUMP-CP CRISPR genes. The pre-registration predicted a positive correlation; the observed relationship is negative (partial $\rho = -0.332$ controlling for n_{contexts}). Genes essential across many cell lines show low DI, and selectively essential or non-essential genes show higher DI. A post-hoc reading is that functional constraint makes downstream phenotypic consequences uniform across contexts; the association attenuates to $\rho = -0.189$ once signal strength is controlled. Line: OLS fit.

tissue-restricted expression (active in few cell types) should show *higher* DI than ubiquitously expressed non-essential genes, because tissue-restricted function implies cell-type-specific downstream consequences. We flag this as a candidate for prospective pre-registration.

The result is consistent with the phenotypic landscape of essential genes being structured by functional constraint [9, 12]. The two morphological atlases cited here do not test cross-cell-type uniformity of phenotypic *direction*: Funk et al. [7] is an optical pooled screen in a single HeLa line, and Ramezani et al. [15] maps morphology across A549 and HeLa without comparing the direction of a gene’s phenotype between them. Neither supports a claim about uniform direction across cell types, and none is made here. DepMap pan-essentiality is likewise uniformity of a scalar viability measure, not of direction in a multivariate phenotype; Experiment H (Section 2.4) shows what happens when the two are conflated.

2.3.2 Knockout against overexpression

If context-dependence were an intrinsic property of genes, then a gene’s knockout DI and its overexpression DI should correlate. Spearman $\rho = 0.011$ (95% CI $[-0.016, 0.038]$, $p = 0.43$, $n = 5,220$ genes with ≥ 5 replicates in both CRISPR and ORF). A mandatory permutation null (1,000 label shuffles preserving plate structure; Appendix C) yielded a null distribution with mean = 0.0003,

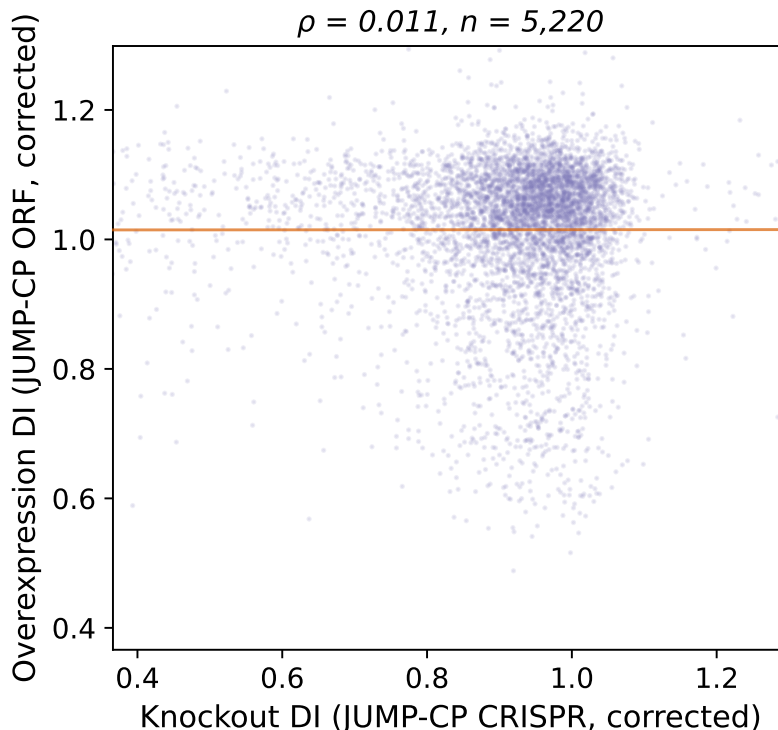


Figure 4: Knockout DI (JUMP-CP CRISPR) vs. overexpression DI (JUMP-CP ORF) for $n = 5,220$ genes profiled under both perturbation types. Spearman $\rho = 0.011$ ($p = 0.43$): no relationship. A gene’s knockout DI does not predict its overexpression DI, so DI does not behave as a stable property of the target gene. Line: OLS fit, slope indistinguishable from zero.

SD = 0.014, 95th percentile = 0.023. The observed $\rho = 0.011$ falls within the null (permutation $p = 0.215$).

At $n = 5,220$, the study had power to detect effects as small as $\rho = 0.04$ (SE ≈ 0.014). The null is well-powered: losing a protein and gaining excess protein engage different molecular mechanisms—loss-of-function versus gain-of-function—and those mechanisms interact with cellular context independently [1, 3].

Figure 4 shows the absence of any relationship. This result constrains the theoretical scope of DI. A gene is context-dependent in a particular way (knockout, overexpression, pharmacological inhibition), and DI values from one perturbation type carry no information about another. A DI atlas must be stratified by perturbation mechanism.

2.3.3 Expression variance, paralog count, paralog breadth

Three plausible proxies for context-dependence—expression variance across cell lines, paralog count, and paralog expression breadth—were specified in the second analysis plan. All three fail after confound control, and the failures are informative about the structure of the confounds.

Expression variance. Partial Spearman $\rho = 0.060$ (95% CI [0.038, 0.082], $p = 1.1 \times 10^{-7}$, $n = 7,802$), controlling for n_{contexts} . The correlation is statistically unambiguous but biologically negligible: $\rho = 0.060$ explains less than 0.4% of variance. The CI lower bound (0.038) does

not clear the 0.05 floor. Expression variance captures a real but negligible relationship: genes with variable expression produce marginally more variable knockout phenotypes. The remaining DI variation reflects factors beyond expression variance. Expression data from DepMap 24Q4 (OmicsExpressionProteinCodingGenesTPMLogp1.csv, 1,673 cell lines, 19,193 genes).

Paralog count. Partial Spearman $\rho = -0.007$ (95% CI $[-0.031, 0.017]$, $p = 0.57$, $n = 6,894$), controlling for n_{contexts} and mean expression level. The raw Spearman correlation was $\rho = 0.038$ ($p = 0.002$)—a statistically significant positive association. The partial correlation is zero. The raw association existed because highly expressed genes have more paralogs (gene family expansion correlates with expression level) and independently have higher DI. Controlling for expression removes the confound entirely. The significant raw association ($p = 0.002$) evaporates under covariate control. Paralog data from Ensembl BioMart (GRCh38, Ensembl 113).

Paralog expression breadth. Among genes with ≥ 1 expressed paralog ($n = 6,882$), partial Spearman $\rho = -0.029$ (95% CI $[-0.053, -0.005]$, $p = 0.017$), controlling for n_{contexts} and mean expression level. The sign is correct (broader paralog compensation predicts lower DI, as hypothesized), but $p = 0.017$ exceeds the Bonferroni threshold ($\alpha = 0.0167$) and the CI upper bound (-0.005) does not clear -0.05 . Both criteria fail: neither p -value nor CI gate passes. The raw correlation ($\rho = -0.117$, $p \sim 10^{-22}$) is substantially larger, indicating that expression-level confounds inflate the apparent paralog-breadth effect as well.

2.3.4 Drug selectivity

Partial Spearman $\rho = 0.144$ (95% CI $[-0.088, 0.360]$, $p = 0.22$, $n = 75$), controlling for LINCS n_{contexts} . The primary metric is magnitude-corrected Gini of PRISM kill depth, residualizing raw Gini against mean kill magnitude to remove the mechanical confound between potency and selectivity. The correction reduced the apparent association from $\rho = 0.275$ (raw Gini) to $\rho = 0.144$ (corrected Gini), confirming the magnitude confound is real.

The CI spans $[-0.088, 0.360]$, consistent with anything from no effect to a moderate one. The pre-registration assumed $n \sim 1,000$; actual overlap between LINCS drugs and PRISM compounds with $|\text{mean LFC}| \geq 0.01$ is $n = 75$, an order of magnitude smaller. This experiment cannot distinguish a real $\rho = 0.15$ effect from zero.

2.4 Replication and construct divergence

2.4.1 Essentiality across two cell lines

The essentiality sign-flip reported above ($\rho = -0.33$, $n = 7,653$) was computed on plate-replicate DI within a single cell line (U2OS). Plate replicates measure technical reproducibility, and reviewers asked whether the sign-flip reflects biology or plate-level noise.

The cpg0000 pilot dataset (Cell Painting Gallery) provides CRISPR profiles from two cell lines: A549 (10 plates) and U2OS (8 plates). With $K = 2$ cell lines, DI degenerates to a single pairwise cosine distance: $\text{DI} = 1 - \cos(\mathbf{s}_{\text{A549}}, \mathbf{s}_{\text{U2OS}})$. This is geometrically valid but extremely noisy per gene (no averaging over pairs). Cell line assignments were obtained from the companion metadata repository [3].

Of 128 genes with profiles in both cell lines, Spearman $\rho = -0.264$ (95% CI $[-0.418, -0.095]$, $p = 0.003$, $n = 128$). The correlation is negative, statistically significant at $\alpha = 0.0167$, and the

CI excludes zero. The pre-registered verdict is **directional replication**: the essentiality sign-flip observed on plate replicates holds in direction on true cross-cell-line data, giving provisional directional evidence against a purely technical reading to biological context-dependence.

The point estimate ($\rho = -0.26$) is smaller than the plate-replicate result ($\rho = -0.33$), consistent with the expected attenuation from $K = 2$ noise. The CI is wide ($[-0.42, -0.10]$), reflecting the small sample size ($n = 128$ against 7,653) and the degeneracy of single-cosine DI. This experiment narrows the interpretation of the sign-flip; it does not close it. Properly powered cross-cell-line replication requires datasets with $K \geq 5$ cell lines per gene, such as future expansions of JUMP or single-cell CRISPR screens [21].

2.4.2 Scalar dependency variability

The biological interpretation of the essentiality sign-flip—functional constraint suppresses context-dependent phenotypes—predicts that the same pattern should appear in dependency scores: broadly essential genes should show uniform dependency across cell lines (low variability), while selectively essential genes should show variable dependency (high variability). Testing this on DepMap Chronos scores provides convergent evidence from a fundamentally different measurement.

Partial Spearman $\rho = +0.811$ (95% CI $[0.806, 0.816]$, $p \approx 0$, $n = 17,916$), controlling for the number of non-missing Chronos values per gene. The pre-registered prediction was negative ($\rho < -0.05$); the observed correlation is strongly positive. This test fails.

We hypothesize the failure reflects a construct divergence rather than a biological contradiction. DepMap essentiality breadth (fraction of cell lines with Chronos < -0.5) and dependency-score SD are different geometric constructs from morphological DI. Morphological DI measures pairwise cosine distance between high-dimensional vectors; dependency-score SD measures the spread of scalar values. A gene essential in many cell lines has a wider range of Chronos scores (some near -1 , others near -0.5), producing high SD. The positive correlation may reflect this scalar-range property: broadly essential genes show high dependency-score variability because the Chronos scale admits more spread as more cell lines cross the essentiality threshold. Consistent with this interpretation, dependency-score SD correlates with Chronos score range at $\rho = 0.83$: SD is largely a proxy for the mechanical range of scores a gene spans, which widens as more cell lines cross the essentiality threshold ($\rho = 0.79$ between score range and breadth).

The failure exposes a construct divergence: the functional-constraint hypothesis makes different predictions depending on whether context-dependence is measured as vector cosine distance (morphological DI) or scalar variability (dependency-score SD). The morphological sign-flip stands; the convergent pathway through DepMap SD does not replicate it. A convergent test using a vector-valued DepMap representation (such as gene-expression signatures of knockout effects across cell lines) would avoid this scalar artifact.

3 Discussion

3.1 What DI measures—and what it does not

Eleven pre-registered experiments produce a consistent characterization of direction instability. DI carries a modest cross-modal component, evidence that it is not specific to one measurement platform. It is perturbation-type-specific: knockout and overexpression of the same gene produce unrelated DI profiles, so DI does not behave as a stable property of the target gene. Broad essentiality is associated with lower DI after measured signal-strength adjustment; functional constraint producing stereotyped disruption is one hypothesis that would account for this, and the association

directionally replicates on true cross-cell-line CRISPR data ($\rho = -0.26$, cpg0000 pilot). Three plausible proxy explanations—expression variance, paralog count, paralog breadth—each explain less than 0.4% of DI variance after confound control.

DI is also mechanism-structured at the pharmacological level: drug MoA classes explain 20% of residualized DI variance, with an interpretable pattern distinguishing core-machinery inhibitors (low DI) from receptor-signaling drugs (high DI). Cross-context consistency is complementary to selectivity: a highly selective drug can still show high DI if its target pathway is wired differently across cell types.

The DepMap convergent test exposes a boundary of the functional-constraint interpretation. Dependency-score SD and essentiality breadth correlate positively ($\rho = +0.81$), opposite to the morphological sign-flip, because the scalar SD estimand mechanically inflates with the Chronos score range. The functional-constraint hypothesis makes different predictions depending on whether context-dependence is measured as vector cosine distance or scalar variability. The morphological sign-flip is about directional consistency of high-dimensional profiles; DepMap SD indexes scalar spread. Future convergent tests should use vector-valued representations of dependency effects.

This combination of positive, negative, and failed results shows that DI captures how perturbation mechanisms interact with cell-type-specific biology. It is not reducible to essentiality breadth, expression variance, or paralog structure, though the association with essentiality breadth survives adjustment at $\rho = -0.189$. Other untested features such as protein interaction network topology or chromatin accessibility might correlate with DI; the present results constrain the space of explanations without exhausting it. For atlas users, the practical implication is that DI provides information beyond what is available from essentiality screens, expression atlases, or paralog databases.

3.2 The resource

The atlas provides magnitude-corrected DI values with bootstrap confidence intervals for 8,949 LINCS drugs, 379 Tahoe drugs, 7,946 CRISPR knockouts, and 12,590 ORF overexpressions. The companion JUMP-CP compound DI values (25,254 compounds) are available from a prior study [22]. Together, these cover the perturbation-biology datasets most widely used for drug mechanism prediction, gene function annotation, and virtual screening benchmarks.

The atlas is designed for downstream integration. Each row provides the columns needed for a secondary analysis: corrected DI, raw DI, CI bounds, mean signature norm, magnitude CV, and number of contexts. Pre-registration documents, analysis scripts, and result JSON files are released alongside the data.

3.3 Limitations

Statistical power. The PRISM viability analysis ($\rho = 0.359$, $n = 75$) is exploratory: uncorrected p -value, no covariate adjustment, small sample. It provides corroboration only. The same-modality concordance test (LINCS-Tahoe, $n = 76$) was underpowered by design: the LINCS-Tahoe compound overlap is structurally small. The point estimate ($\rho = 0.254$) is consistent with genuine concordance, but the wide CI ($[0.027, 0.456]$) precludes a firm conclusion. The drug selectivity test is underpowered at $n = 75$ and cannot distinguish a moderate effect from zero. Both underpowered tests are data-limited: the small n reflects the structural size of the dataset intersection, not a design choice. We report them for completeness and for the atlas resource record, not as adjudicated hypotheses. Experiment G computes DI from $K = 2$ cell lines, the minimum for a pairwise comparison. With $K = 2$, DI is a single cosine distance rather than an average over pairs,

making it extremely noisy per gene. No magnitude correction was applied ($n = 128$ is too small for meaningful OLS), and no bootstrap CIs are possible. The A549 plates vary in timepoint (96h vs. 144h) and antibiotic treatment, introducing uncontrolled confounds. The experiment’s value is as a directional check, not a definitive test.

Construct and interpretation. Context definitions vary across datasets. LINCS and Tahoe use cell lines as contexts; JUMP-CP CRISPR and ORF use plate replicates. Plate replicates sample technical variability within one cell line, while cell lines sample biological variability across cell types. DI computed over plate replicates measures reproducibility within a fixed biological context, whereas DI over cell lines measures context-dependence across biological contexts. Cross-dataset comparisons match on context type, but the CRISPR/ORF DI values used in the essentiality, mechanism-specificity, and proxy tests index technical variability rather than biological context-dependence. Extending the atlas to single-cell CRISPR screens with cell-line diversity [21] would resolve this asymmetry. The essentiality sign-flip interpretation (functional constraint suppresses DI) is post-hoc. The pre-registration predicted the opposite sign. The mechanistic explanation is plausible and connects to results in functional genomics [7, 15], but it was generated after observing the data. A companion preprint by the same author reports that universal-essential knockdowns diverge rather than converge across cell types in Perturb-seq data ($d = 0.60$, $p_{\text{perm}} < 10^{-4}$; 23); that finding is not independent of the present work, and its direction is opposite to the uniform-disruption reading, so it is recorded here as an open tension rather than as support. Replogle et al. [18] bear directly on the question: genome-scale Perturb-seq in K562 and RPE1 finds only partial concordance of knockdown phenotypes across the two lines, which is consistent with context-dependence being widespread and argues against reading low DI as evidence of uniform cross-cell-type disruption. Prospective testing of a falsifiable prediction—among non-essential genes, tissue-restricted expression should positively predict DI (Section 2.3)—would convert this post-hoc rationalization into a validated mechanism. Experiment F is exploratory-then-confirmed (scoping disclosure, Section 2.4); the confound-controlled $\eta^2 = 0.20$ is the primary result.

Experiment H exposes a construct divergence: scalar dependency-score SD and vector cosine DI measure different geometric properties of context-dependence. The positive $\rho = +0.81$ does not falsify the morphological sign-flip; it shows that the functional-constraint hypothesis generates different predictions under different operationalizations.

Data scope. LINCS L1000 has well-documented batch effects across production sites and plates [17]. If a drug was profiled across batches, batch-driven direction differences would inflate DI independently of biology. The companion study’s split-half stability analysis provides indirect reassurance, but a direct test of whether DI correlates with batch structure has not been performed. All DI values are computed at the dose used in each screening campaign. Dose-response effects could change a drug’s DI: a compound selective at low dose might become promiscuous at high dose, altering its context-dependence. The atlas does not capture this dose-dependent structure.

3.4 Future directions

Four extensions follow naturally from the current results. First, Batch 2B will test whether DI correlates with Shesha geometric stability [13, 14], a concurrent and independently developed measure of perturbation coherence computed on single-cell data. Second, Gene Ontology enrichment analysis on the directional divergence gene sets from the mechanism-specificity test (1,331 knockout-stable / overexpression-unstable genes; 289 with the opposite pattern) may reveal functional categories associated with mechanism-specific context-dependence. Third, expanding the Tahoe drug-InChIKey

annotation would provide a properly powered same-modality concordance test. Fourth, a properly powered cross-cell-line CRISPR replication ($K \geq 5$ cell lines per gene) would resolve the essentiality sign-flip interpretation definitively; future expansions of JUMP or single-cell CRISPR screens [21] may provide such data.

More broadly, the failure of tested gene-level annotations to explain DI, combined with the success of pharmacological MoA class in structuring it, suggests that DI indexes an aspect of perturbation biology—the interaction between mechanism and cellular context—that gene-level annotations do not capture. Whether this interaction can be predicted from protein interaction networks, pathway topology, or chromatin accessibility remains an open question.

3.5 Conclusions

Eleven pre-registered experiments across three batches characterize direction instability as a metric that is concordant across measurement modalities, perturbation-type-specific, associated with essentiality breadth, and structured by pharmacological mechanism-of-action class. DI is not reducible to expression variance, paralog count, or paralog breadth. The essentiality sign-flip, for which functional constraint is one candidate explanation, directionally replicates on cross-cell-line CRISPR data. A convergent test on DepMap dependency-score variability fails, exposing a construct divergence between scalar SD and vector cosine DI that bounds the generality of the functional-constraint interpretation. The atlas provides magnitude-corrected DI values with bootstrap confidence intervals for 8,949 drugs, 379 single-cell-profiled drugs, 7,946 CRISPR knockouts, and 12,590 ORF overexpressions, ready for integration with drug-mechanism prediction, gene-function annotation, and screening quality control.

Data and code availability

All analysis scripts are written in Python 3 with standard scientific libraries (NumPy, SciPy, pandas, scikit-learn) and are available at <https://github.com/elliotttower/direction-instability-atlas> under the MIT License.

All atlas data files (per-perturbation DI values with bootstrap CIs for the Tahoe-100M, CRISPR knockout, and ORF overexpression datasets, together with PRISM viability summaries), experiment result files, pre-registration documents, and analysis code are deposited on Zenodo (DOI: 10.5281/zenodo.21223667, first deposited 6 July 2026) under a CC-BY-4.0 license. This manuscript was first submitted for peer review on 6 July 2026 and the companion study on 5 July 2026; both dates precede the concurrent 2026 work discussed in Section 3, and are given here so priority can be checked rather than assumed. Pre-registration commits can be verified against the repository history at the SHA values reported in the text. LINCS L1000 compound DI values (8,949 drugs) and JUMP-CP compound DI values (25,254 compounds) were previously computed and are available from the companion study (DOI: 10.5281/zenodo.21213699).

Source datasets are available from their original repositories: LINCS L1000 (GEO: GSE92742), Tahoe-100M (HuggingFace: tahoebio/Tahoe-100M), JUMP Cell Painting (Cell Painting Gallery, cpg0016), DepMap 24Q4 (figshare: 27993248), PRISM (figshare: Repurposing Public 24Q2), Ensembl paralogs (BioMart, GRCh38).

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Author contributions

E.T.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of interests

The author declares no competing interests. This work was conducted independently with no external funding.

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A Pre-registration protocol summary

All hypotheses, decision criteria, statistical methods, and analysis code were frozen before execution. The three pre-registration documents are included in the Zenodo deposit.

Table 4: Pre-registration batches.

Batch	Commit SHA	Experiments
Batch 1	0d07a01	A2 (LINCS-JUMP cross-modal concordance), A1 (LINCS-Tahoe same-modality concordance), C (essentiality), D (KO vs. OE), PRISM (exploratory)
Batch 2A	a17c125	E2 (expression variance), E3 (drug selectivity), E5a (paralog count), E5b (paralog breadth)
Batch 3	4c34699	F (MoA stratification), G (cross-cell-line CRISPR essentiality), H (DepMap convergent evidence)

Two Batch 1 experiments were dropped before analysis: A3 (Tahoe-JUMP concordance; zero compound overlap between clinical drugs and screening library) and B (MOA stratification in Tahoe; structurally underpowered at $n = 7$ vs. $n = 17$). Batch 2A labels E2/E3/E5 are non-consecutive because the pre-registration proposed three specific explanatory hypotheses, not a sequential series. Batch 3 was prompted by peer review and frozen at a separate SHA with its own $\alpha = 0.0167$ (three tests).

What did not change across batches. The DI computation (Equation 1), the magnitude correction (Equation 2), the partial Spearman method (Pearson-residualize then rank), the CI floor gate (± 0.05), and the Bonferroni correction framework were constant across all three batches. Batch 2A and Batch 3 each added hypotheses with their own $\alpha = 0.0167$ (Bonferroni for three tests each).

B PRISM viability as a scalar check

As an independent check on construct validity, we computed the Spearman correlation between LINCS DI and PRISM viability CV for $n = 75$ shared drugs (filtered to $|\text{mean LFC}| \geq 0.01$). Spearman $\rho = 0.359$ (95% CI [0.144, 0.542], $p = 0.0016$).

This result is **exploratory**: n is small, the p -value is uncorrected, and no covariate adjustment was applied. It should carry weight only alongside the primary tests. That said, viability is a scalar outcome with a fundamentally different noise structure from high-dimensional expression or morphology profiles. The correlation suggests that compounds whose transcriptomic effects depend on cellular context also show variable cell-killing potency, consistent with the interpretation that DI captures a biological property of drug-context interactions.

C Knockout vs. overexpression permutation null

The mandatory permutation null for the knockout vs. overexpression DI correlation used 1,000 label shuffles preserving plate structure within each dataset. The null distribution had mean = 0.0003, SD = 0.014, and 95th percentile = 0.023. The observed $\rho = 0.011$ falls at permutation $p = 0.215$,

well within the null. At $n = 5,220$, the study had power to detect effects as small as $\rho = 0.04$ ($SE \approx 0.014$), so the null result is well-powered.

D Same-modality concordance matching method comparison

The pre-registration specified InChIKey matching for drug concordance. An initial run erroneously used case-insensitive drug-name matching, which inflated the sample size and introduced spurious pairs.

Table 5: Same-modality concordance results by matching method. InChIKey matching is the pre-registered analysis.

Method	Match type	n	Partial ρ	Note
Drug-name (erroneous)	Case-insensitive	114	-0.003	~38 spurious pairs from salt forms, stereoisomers
InChIKey (pre-registered)	Connectivity prefix	76	0.254	Via PubChem CID cross-reference

Reverting to InChIKey matching resolved the spurious null. The name-matched result ($\rho = -0.003$) reflects contamination by chemically distinct compounds sharing common drug names, not absence of concordance.

E Magnitude correction details

The OLS magnitude correction (Equation 2) is applied independently per dataset. For each dataset, raw DI is regressed on mean signature L_2 norm. The residuals plus the intercept yield $DI_{\text{corrected}}$.

Table 6: Magnitude correction parameters per dataset. Negative slopes indicate that weaker perturbations (lower norm) have higher raw DI, as expected from the noise-inflation mechanism.

Dataset	n	OLS slope	R^2
Tahoe-100M	379	+0.00013	0.002
JUMP-CP CRISPR	7,946	-0.0097	0.081
JUMP-CP ORF	12,590	-0.0126	0.182

LINCS L1000 and JUMP-CP compound DI values were corrected in the companion study [22] and are distributed with pre-corrected values. The near-zero Tahoe slope reflects Tahoe-100M’s high signal-to-noise ratio (deeply sequenced single-cell data); the correction has negligible effect on this dataset.